Compute-Efficient Generative Question Answering

LLM Semester Project • EN.737 • Five-Person Team

1. Project Idea

This project investigates training, generalization, and inference efficiency in small generative language models under a limited compute budget.

We will adapt **Qwen3-0.6B** to context-grounded, single-turn question answering using **LoRA supervised fine-tuning**, starting from zero-shot and few-shot prompting baselines.

We will use standardized **MRQA** datasets, including SQuAD, NewsQA, and TriviaQA for training, and BioASQ and DuoRC for held-out generalization evaluation.

Each input consists of a context passage and a question. The expected output is a concise answer supported by the passage. Primary quality metrics are Exact Match (EM) and token-level F1.

2. Research Questions

RQ1 — Training Efficiency

How does QA performance change with the LoRA training-token budget?

Compare prompting baselines with different fine-tuning budgets using learning curves, validation loss, EM/F1, and GPU cost.

RQ2 — Generalization

Does multi-source fine-tuning improve out-of-distribution (OOD) generalization under a fixed training-token budget?

Compare SQuAD-only training against SQuAD + NewsQA + TriviaQA, holding the base model, LoRA configuration, and total training tokens constant. Evaluate on unseen BioASQ and DuoRC questions using EM/F1, paired bootstrap confidence intervals, and error analysis.

RQ3 — Inference Efficiency

How does inference batch size affect throughput, latency, and GPU memory consumption?

Evaluate the same fine-tuned model under controlled inference configurations. KV-cache experiments are optional.

3. Team Responsibilities

Member	Responsibility
P1 — Data & Evaluation	Dataset preparation and validation, train/dev/test splits, data-overlap checks, EM/F1 evaluation pipeline, prediction logging.
P2 — Modeling & Training	Prompting baselines, shared Qwen-LoRA training pipeline, training-budget experiments, loss and computational-cost analysis.
P3 — Generalization	Design and run single-source versus multi-source fine-tuning experiments, ID/OOD evaluation, statistical comparisons, and error analysis.
P4 — Inference Efficiency	Batching experiments, latency and throughput measurement, GPU memory profiling, quality-efficiency tradeoffs.
P5 — Tiny Transformer	Implement and train a decoder-only Transformer from scratch; analyze cross-entropy, parameter counts, training-token budgets, and limitations.

4. Current Status and Deliverables

Preliminary prompting experiments and a 1,000-example Qwen LoRA pilot have demonstrated technical feasibility. Formal controlled experiments remain to be conducted.

Before the main experiments, we will establish hypotheses and quantitative predictions, fix experimental configurations, repeat key comparisons, and report statistical uncertainty.

Final deliverables: Reproducible Colab notebooks or GitHub repository, experiment configurations, saved predictions, evaluation results, a short research report, and a team contribution statement.

Each member will own a substantive technical contribution, while final report preparation and interpretation will be shared by the team.